A

MINI PROJECT REPORT

ON

IMAGE TO PENCIL SKETCH

submitted in partial fulfilment of the required for the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING

(Session 2025-26)



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DECLARATION

I am KHUSHI GUPTA (2408390109003) who carried out the project entitled “IMAGE TO PENCIL SKETCH" done by us under the guidance of Prof. Dr. B.D.K. Patro sir is submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in Computer Science Engineering. To the best of our knowledge this project has not been submitted to Dr. A. P.J. Abdul Kalam Technical University Lucknow, or any other University or Institute for the award of any degree.

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CERTIFICATE

This is to certify that this Project Report is the Bonafide work of KHUSHI GUPTA (2408390109003) who carried out the project entitled “IMAGE TO PENCIL SKETCH” This project has been undertaken in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE ENGINEERING at RAJKIYA ENGINEERING COLLEGE, KANNAUJ. The project work demonstrates their dedication, teamwork, and commitment to achieving technical excellence and contributes to the development of sustainable energy solutions. We commend their efforts and wish them success in their future endeavors.

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ACKNOWLEDGEMENT

We are pleased to acknowledge our sincere thanks to Board of Management of RAJKIYA ENGINEERING COLLEGE for their kind encouragement in doing this project and for completing it successfully. We are grateful to them. We convey our thanks to Dean, Department of Computer Science Engineering and Head of the Department, Dept. of Computer Science Engineering for providing us necessary support and details at the right time during the progressive reviews. We would like to express our sincere and deep sense of gratitude to our Project Guide Dr. B. D. K. Patro Sir for his valuable guidance, suggestions and constant encouragement paved way for the successful completion of our project work. We wish to express our thanks to all Teaching and Non-teaching staff members of the Department of Computer Science Engineering who were helpful in many ways for the completion of the project.

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ABSTRACT

The "Image to Pencil Sketch" project explores the digital transformation of standard photographs into artistic, hand-drawn renderings through advanced image processing techniques. This project implements a comprehensive pipeline that bridges the gap between digital photography and fine art by emphasizing structural elements like contours and shading while removing unnecessary color data.

The methodology utilizes a systematic multi-step strategy involving Grayscale Conversion, Image Inversion, Gaussian Blur for texture simulation, and final Edge Detection and Blending. While classical methods provide an efficient and computationally inexpensive way to generate sketches, the project also examines the integration of modern AI, such as Convolutional Neural Networks (CNNs) and Neural Style Transfer, to overcome traditional limitations like "flat" appearances and repetitive textures.

Beyond simple aesthetic filters, this report details the professional utility of sketch conversion in fields such as Architecture (Sketch-to-BIM), Industrial Design, and Immersive Media (AR/VR). Ultimately, the project demonstrates how the synergy between traditional image processing and emerging AI technologies can create sophisticated tools for both creative expression and professional technical documentation.

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Chapter 1

Introduction

1.1 About Image to Pencil Sketch

Image to Pencil Sketch is a digital image processing technique that converts a normal color or grayscale photograph into a pencil-drawn style image. The goal of this transformation is to mimic the appearance of a hand-drawn sketch by emphasizing edges, contours, and shading while suppressing unnecessary color information.

This process is widely used in computer vision and graphics applications such as photo editing, artistic rendering, animation, and mobile camera filters. Traditionally, image-to-sketch conversion is achieved using classical image processing methods like grayscale conversion, edge detection, image inversion, Gaussian blur, and image blending. These techniques simulate the texture and strokes of a pencil drawing in an efficient and computationally inexpensive way.

With advancements in artificial intelligence, deep learning–based approaches such as convolutional neural networks (CNNs) and neural style transfer have also been applied to generate more realistic and high-quality pencil sketches. These methods learn artistic features directly from data and can produce sketches that closely resemble human drawings.

Overall, image to pencil sketch conversion bridges the gap between photography and art, providing an automated way to create visually appealing sketches from digital images.

1.2 Types of Image to Pencil Sketch

The types of Image to Pencil Sketch methods can be broadly classified into the following categories:

1. Classical Image Processing–Based Sketch

- This method uses traditional image processing techniques such as grayscale conversion, image inversion, Gaussian blur, edge detection (Sobel, Canny), and image blending.
- It is simple, fast, and does not require training data, making it suitable for basic sketch effects.

2. Edge Detection–Based Sketch

- In this approach, the sketch is created mainly by detecting object edges and contours in the image.
- Operators like Canny, Laplacian, or Sobel highlight strong edges, producing a line-drawing or outline-style pencil sketch.

3. Thresholding and Morphological Sketch

- This type uses thresholding and morphological operations to extract important structures and create bold, high-contrast sketch images.
- It is often used for cartoon-style or rough pencil sketches.

4. Machine Learning–Based Sketch

- These methods use machine learning algorithms trained on image–sketch pairs to learn how to convert photos into sketches.
- They produce better results than traditional methods but require labeled datasets.

5. Deep Learning–Based Sketch

- Deep learning techniques such as Convolutional Neural Networks (CNNs), Autoencoders, and Neural Style Transfer generate highly realistic pencil sketches.
- These models learn artistic features and shading patterns from large datasets and provide high-quality outputs.

6. Real-Time Filter–Based Sketch

- This type is commonly used in mobile apps and cameras.
- It applies optimized image processing or lightweight neural networks to generate pencil sketches instantly in real time.

1.3 Image Processing

Image processing is a method to perform operations on an image to enhance it or extract useful information from it. It is a type of signal processing in which the input is an image (like a photograph or video frame) and the output may be an image or characteristics/features associated with that image.

1. How Image Processing Works

The process typically follows a series of steps, starting from the moment a sensor captures light to the final output.

- **Image Acquisition:** Capturing an image through a sensor (like a camera) and converting it into a digital format (pixels).
- **Image Enhancement:** Improving the visual quality by adjusting brightness, contrast, or sharpness to make it more suitable for a specific task.
- **Image Restoration:** Using mathematical models to "undo" damage like blurriness, noise, or camera shakes.
- **Color Image Processing:** Handling color models (like RGB or HSV) to improve vibrancy or perform color-based detection.
- **Segmentation:** Partitioning the image into multiple segments (objects or regions) to make it easier to analyze.
- **Recognition:** Assigning a label to an object based on its descriptors (e.g., identifying a "car" vs. a "pedestrian").

2. Core Techniques

Modern image processing uses a mix of traditional mathematics and advanced AI:

Technique	Description	Common Use Case
Filtering	Modifying pixels based on their neighbors (e.g., Gaussian Blur).	Noise reduction & smoothing.
Edge Detection	Identifying points where image brightness changes sharply.	Finding boundaries of objects.
Morphological Ops	Processing images based on shapes (Erosion, Dilation).	Removing "dots" or connecting gaps.
Compression	Reducing the file size without losing essential data.	JPEG files, streaming video.

Thresholding	Converting a grayscale image into binary (black and white).	Separating text from a background.

3. Real-World Applications (2025)

Image processing is no longer just about "filters"—it powers the most advanced tech today:

- **Medical Imaging:** Automated detection of tumors in MRI or X-ray scans with higher precision than the human eye.
- **Autonomous Vehicles:** Real-time detection of traffic lights, lane markings, and pedestrians to navigate safely.
- **Biometrics:** Facial recognition in smartphones and airport security.
- **Agriculture:** Drones using multispectral imaging to detect crop diseases or soil hydration levels.
- **E-commerce:** "Visual search" where you take a photo of a product to find it online.

4. Analog vs. Digital

- **Analog Image Processing:** Deals with hard copies like printouts and photographs. It uses optical or electrical means to manipulate images (e.g., a magnifying glass or a darkroom).
- **Digital Image Processing:** Uses computer algorithms to manipulate digital images. This is the dominant form today because it is flexible, precise, and can be automated at scale.

Chapter 2

Methodology and Strategy

2.1 Methodology

The key technologies behind Image to Pencil Sketch

1. Image Preprocessing

Techniques such as resizing, noise reduction, and grayscale conversion prepare the input image for sketch generation by simplifying color information and improving clarity.

2. Edge Detection

Edge detection algorithms like Sobel, Canny, and Laplacian are used to identify object boundaries and contours, which form the basic structure of a pencil sketch.

3. Image Filtering and Blurring

Filters such as Gaussian blur help create smooth shading effects and simulate the soft texture of pencil strokes.

4. Image Inversion and Blending

Image inversion combined with blending techniques (e.g., color dodge) produces the light and dark contrast typical of pencil sketches.

5. Thresholding and Morphological Operations

Thresholding converts images into high-contrast sketches, while morphological operations refine lines and enhance stroke thickness.

6. Computer Vision Libraries

Libraries like OpenCV provide efficient tools for implementing image-to-sketch algorithms and real-time processing.

7. Machine Learning Techniques

Traditional machine learning models learn sketch patterns from image–sketch pairs to improve realism compared to rule-based methods.

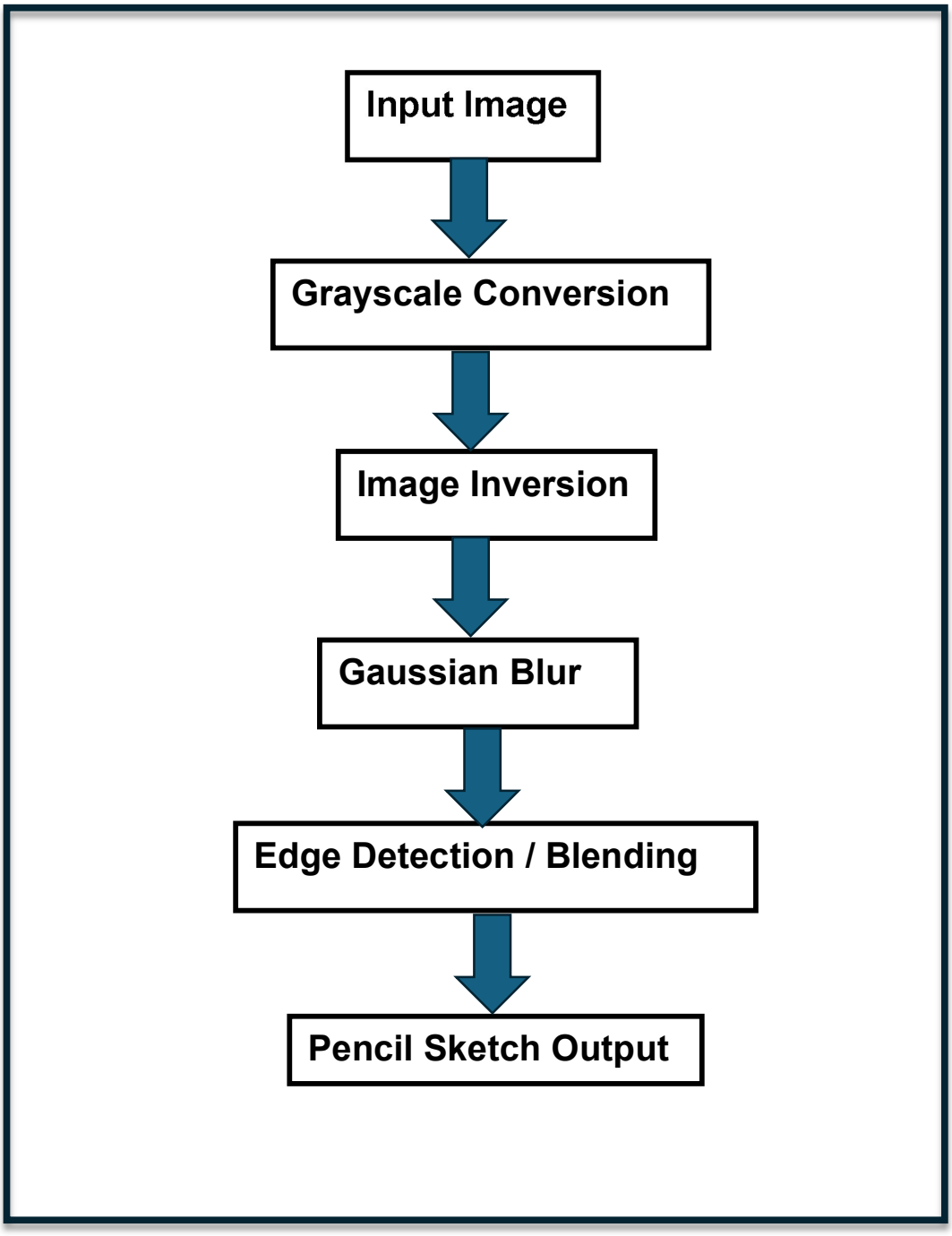
8. Deep Learning and Neural Networks

Advanced models such as Convolutional Neural Networks (CNNs), Autoencoders, and Neural Style Transfer learn artistic features and generate high-quality, realistic pencil sketches.

9. GPU Acceleration

Graphics Processing Units (GPUs) speed up computation, especially for deep learning–based sketch generation and real-time applications.

2.2 Strategy



Chapter 3

Advantages and Disadvantages

3.1 Advantages

1. Aesthetic & Artistic Value

- **Timeless Appeal:** Sketches evoke a classic, "hand-drawn" feel that feels more personal and less clinical than a high-resolution digital photo.
- **Nostalgia & Emotion:** The grayscale and grainy texture of a pencil sketch often create a sense of mood and sentimentality, making them popular for portraits and wedding photos.
- **Minimalism:** By removing color, you focus the viewer's attention on form, texture, and composition rather than distracting hues.

2. Creative & Professional Utility

- **Storyboarding & Mockups**: Designers use sketch filters to create quick conceptual art or storyboards. It allows clients to focus on the "big idea" rather than getting bogged down in the specific details of a finished photograph.
- **DIY Coloring Pages:** Converting a photo to a high-contrast outline sketch is the easiest way to create custom coloring pages for children or adult relaxation.
- **Unique Branding:** For social media or marketing, sketch-style images stand out in a sea of standard photography (the Von Restorff effect), helping a brand appear more artisanal or creative.

3. Technical Benefits

- **Detail Emphasis:** Sketching naturally emphasizes the "bones" of an image—the contours and edges. This can reveal architectural details or facial expressions that were lost in a busy, colorful background.
- **Fixing Poor Photos:** If a photo has bad lighting, color noise, or distracting backgrounds, converting it to a black-and-white sketch can "mask" these flaws while preserving the essence of the subject.
- **Speed & Accessibility:** AI-driven tools allow anyone to create professional-looking "hand-drawn" art in seconds without needing years of training or expensive art supplies.

4. Educational & Learning Tool

- **Artistic Reference:** For aspiring artists, converting a photo to a sketch helps them study values (light vs. dark) and proportions without the complexity of color theory.
- **Composition Analysis:** Seeing an image reduced to lines helps photographers understand the "leading lines" and structural balance of their shots.

3.2 Disadvantages

1. Lack of Realistic Texture and Depth

- **"Flat" Appearance:** Filters often struggle to replicate 3D volume. Because the software is simply tracing edges on a 2D plane, the result can look "flat" compared to a hand-drawn sketch where an artist uses shading to imply mass and depth.
- **Uniform Stroke Patterns:** Automated tools often use repetitive, "procedural" textures. A real artist varies pencil pressure and direction based on the subject (e.g., curved strokes for a face, straight lines for a building), whereas a filter may apply the same "grain" across the entire image.
- **Unnatural Hair and Skin:** Fine details like hair often become a blurry "mush" or look like "injection-molded plastic" because the algorithm cannot distinguish individual strands from the background.

2. Technical Artifacts

- **Edge "Noise":** High-contrast areas can create weird black dots or "halos" around objects. This is especially common on lips or eyes, where the filter misinterprets small shadows as heavy pencil marks.
- **Sensitivity to Image Quality:** If the original photo is low-resolution or blurry, the sketch will look "muddy" or pixelated. Unlike a human artist who can "fill in the blanks" using logic, a filter just amplifies the existing blur.
- **Lost Details:** Important subtle features—like a slight smile or a glint in the eye—can be completely erased if they don't have enough contrast for the edge-detection algorithm to pick up.

3. Loss of Artistic Intent

- **No "Focal Point":** An artist purposefully leaves some areas rough and details others to guide the viewer's eye. A filter treats every pixel with equal importance, which can result in a cluttered, "busy" image that is hard to look at.
- **Lack of Style:** Most "Photo-to-Sketch" results look identical. They lack the unique "handwriting" or emotional touch that defines a specific artist's style.
- **"Uncanny Valley" Effect:** Sometimes the result is *too* close to the photo but with "off" textures, making it look creepy or "cheap" rather than artistic.

4. Professional & Educational Limitations

- **Stunted Skill Growth:** Relying on filters can prevent beginners from learning how to actually "see" form, light, and shadow. It encourages "copying" rather than "creating."
- **Watermarks & High Costs:** Many "free" apps place large watermarks on the result or hide high-resolution exports behind expensive subscriptions.

Chapter 4

Future Scope

The "Future Scope" in IT can be broken down into four major pillars:

1. Professional Architecture & Industrial Design

Traditionally, architects and engineers manually sketched over photographs. New IT tools are making this bidirectional:

- **Bionic Architecture:** Tools like *Autodesk Forma* and *ArchAI* now use AI to instantly convert site photos into technical pencil drafts that respect real-world physics and building codes.
- **Sketch-to-BIM (Building Information Modeling):** Future software will allow a user to take a photo of a room, convert it to a sketch, and then automatically generate a 3D CAD model from those lines, drastically reducing the time for "as-built" documentation.

2. Advanced Generative AI & Personalization

The next wave of AI (like the **Nano Banana** model and **Midjourney 6.0**) is solving the "flatness" problem mentioned earlier:

- **Identity Preservation:** AI is getting better at maintaining the exact likeness of a person while changing the medium. You can now turn a selfie into a sketch that looks like it was drawn by a *specific* famous artist from history.
- **Layered Decomposition:** Modern models (like *Qwen-Image-Layered*) can now break a photo into multiple transparent layers. This means you can convert just the *subject* into a pencil sketch while keeping the background as a photo, or vice versa, for complex graphic design.

3. AR/VR & Immersive Storytelling

- **Real-time Stylization in AR:** Augmented Reality glasses will soon allow users to see the real world through a "sketch" lens. This is useful for **education** (seeing the skeletal structure of a building) or **maintenance** (sketching out internal wiring overlays on a physical wall).
- **3D Sketching in VR:** Instead of sketching on a 2D screen, IT professionals are moving toward "drawing in the air." Tools like *Gravity Sketch* allow designers to turn 2D image references into 3D wireframes in a virtual workspace.

4. Automation in Entertainment (Comics & Animation)

- **Comic/Manga Production:** AI tools are being developed to help artists convert real-life location photos into high-quality manga backgrounds instantly. This automates the most time-consuming part of comic creation, allowing creators to focus on characters and story.
- **Sketch Animation:** Researchers are working on "sketch-based motion transfer," where you can take a static pencil sketch and use AI to animate it by mapping it to a video of a real person dancing or moving.

Chapter 5

Conclusion

The "Image to Pencil Sketch" project demonstrates the effective transformation of digital photographs into artistic renderings by utilizing a systematic pipeline of image processing techniques. The methodology successfully bridges the gap between photography and fine art by converting color or grayscale images into a pencil-drawn style that emphasizes contours and shading while suppressing unnecessary color information. This is achieved through a multi-step strategy involving grayscale conversion, image inversion, Gaussian blur for smoothing, and final edge detection or blending. By following these steps, the project creates visually appealing sketches that mimic hand-drawn aesthetics in an efficient and computationally inexpensive manner.

While the project highlights significant advantages such as artistic minimalism, the ability to mask flaws in poor-quality photos, and professional utility for storyboarding, it also acknowledges inherent limitations. Automated filters often produce a "flat" appearance and repetitive stroke patterns that lack the depth and emotional "handwriting" of a human artist. Technical artifacts like edge noise and the potential loss of subtle facial details further illustrate the gap between algorithmic processing and manual artistry. However, the integration of advanced technologies like Deep Learning and Convolutional Neural Networks (CNNs) provides a pathway to overcome these challenges by learning artistic features directly from data to produce higher-quality, more realistic outputs.

Looking toward the future, the scope of this field in IT is expanding into professional domains such as architecture, industrial design, and immersive media. Technologies like Bionic Architecture and Sketch-to-BIM are set to revolutionize how site photos are converted into technical drafts and 3D models. Furthermore, the advancement of real-time stylization in Augmented Reality (AR) and 3D sketching in Virtual Reality (VR) indicates that pencil sketch conversion is evolving from a simple creative filter into a functional tool for maintenance, education, and industrial design. Ultimately, this project underscores a future where the synergy between traditional image processing and modern AI creates sophisticated tools for both artistic expression and professional technical documentation.

Chapter 6

References

1. Academic & Research Papers

- **Lu, C., Xu, L., & Jia, J. (2012).** *Combining Sketch and Tone for Pencil Drawing Production.* This is a foundational paper in Non-Photorealistic Rendering (NPR) that introduced the two-stage process of generating stroke and tonal layers separately.
- **Gastal, E. S., & Oliveira, M. M. (2011).** *Domain Transform for Edge-Aware Image and Video Processing.* This paper introduces the domain filter used in OpenCV's built-in cv2.pencilSketch function.
- **Ha, D., & Eck, D. (2017).** *A Neural Representation of Sketch Drawings (Sketch-RNN).* Published by Google Research, this explores how Recurrent Neural Networks (RNNs) can learn to draw vector-based sketches like a human.
- **Gatys, L. A., Ecker, A. S., & Bethge, M. (2016).** *Image Style Transfer Using Convolutional Neural Networks.* A seminal paper on Neural Style Transfer (NST) which paved the way for modern AI-driven sketch filters.

2. Standard Textbooks & Reference Books

- **Gonzalez, R. C., & Woods, R. E. (2018).** *Digital Image Processing (4th ed.).* Pearson. This is the global standard textbook for understanding the mathematical foundations of grayscale conversion, blurring, and edge detection.
- **Howse, J., & Minichino, J. (2020).** *Learning OpenCV 4 Computer Vision with Python 3.* Packt Publishing. Provides hands-on implementation details for Gaussian filters and bitwise operations used in sketching.
- **Beyeler, M. (2015).** *OpenCV with Python Blueprints.* Packt Publishing. Specifically contains chapters on "Fun with Filters" that detail the "Dodge and Burn" blending technique for pencil sketches.

3. Software Libraries & Technical Documentation

- **OpenCV-Python Documentation:** https://docs.opencv.org/ – Specifically for cv2.cvtColor, cv2.GaussianBlur, and cv2.divide functions used in the project pipeline.

- **NumPy Documentation:** https://numpy.org/doc/ – For array manipulations and bitwise operations (e.g., 255 - gray_image) essential for creating the negative image.
- **Matplotlib Documentation:** https://matplotlib.org/ – Used for visualizing the side-by-side comparison of the original photo and the final output.

4. Relevant Projects & Articles

- **Kaggle:** *Image to Pencil Sketch using cv2* by Infinator. A widely cited community project demonstrating the standard 5-step Python sketching pipeline.
- **IEEE Xplore:** *Image to Pencil Sketch Converter using Python* (2023) by Rohan Gupta et al. A recent paper discussing the implementation of an interface for sketch conversion using OpenCV.

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